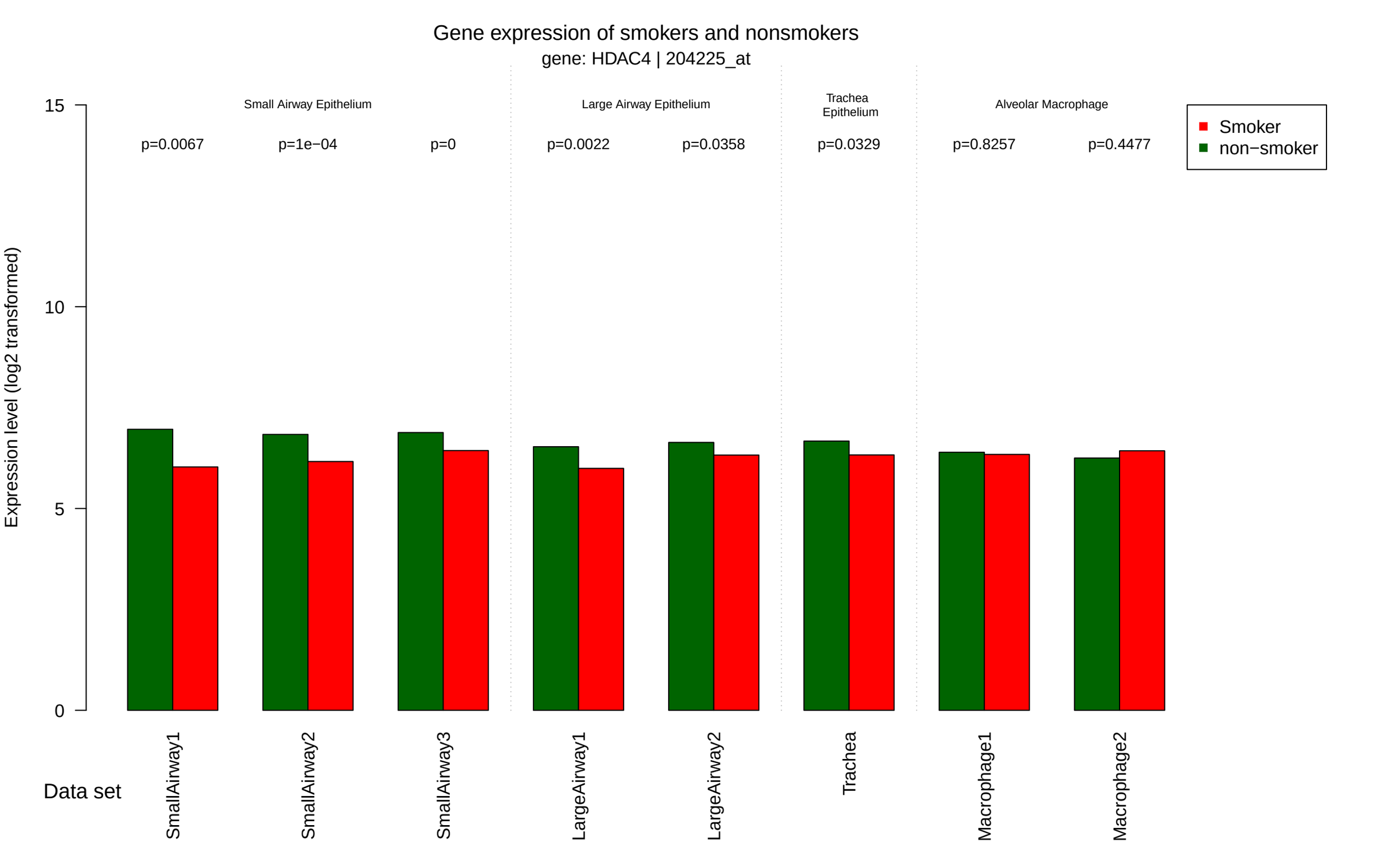
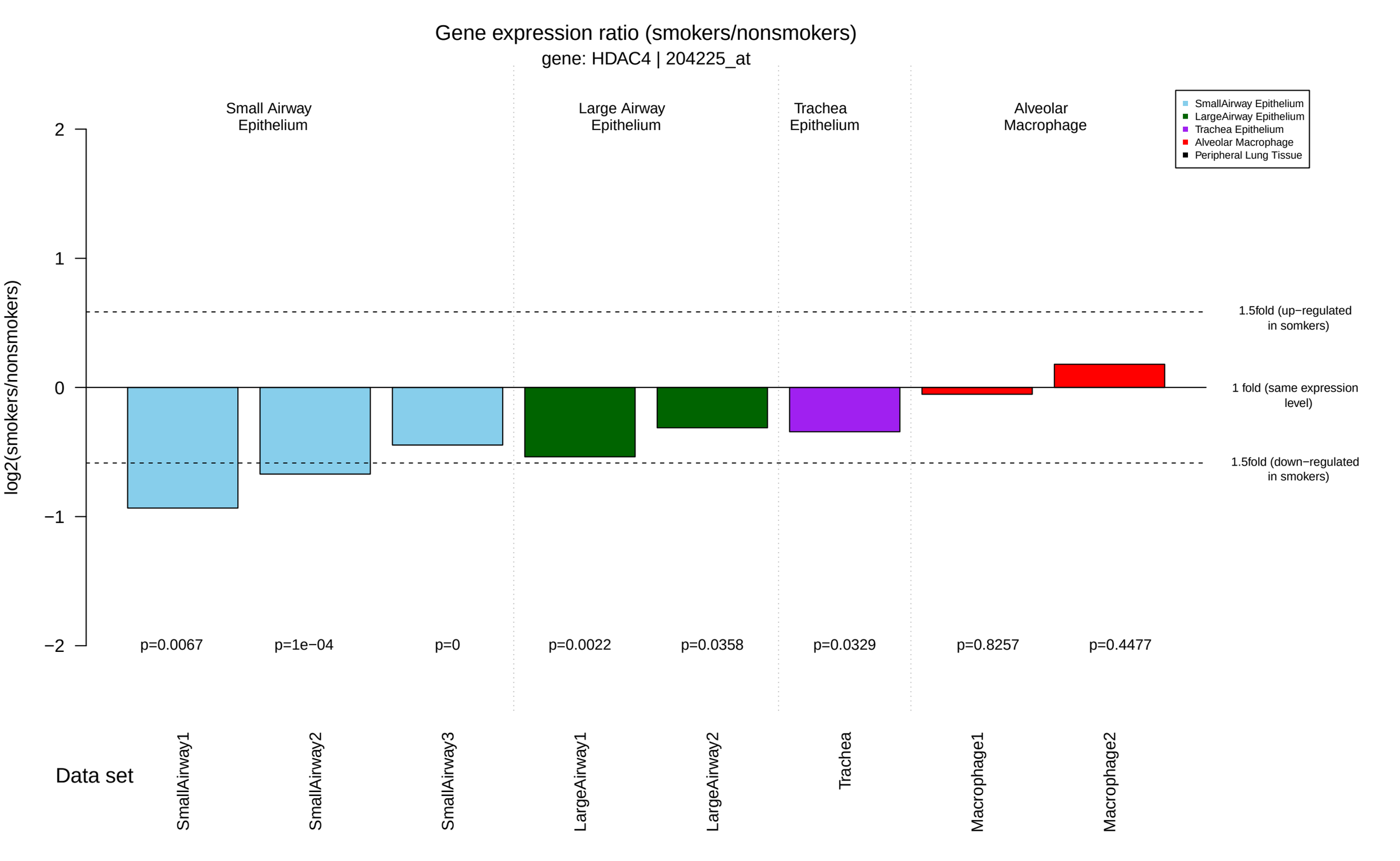
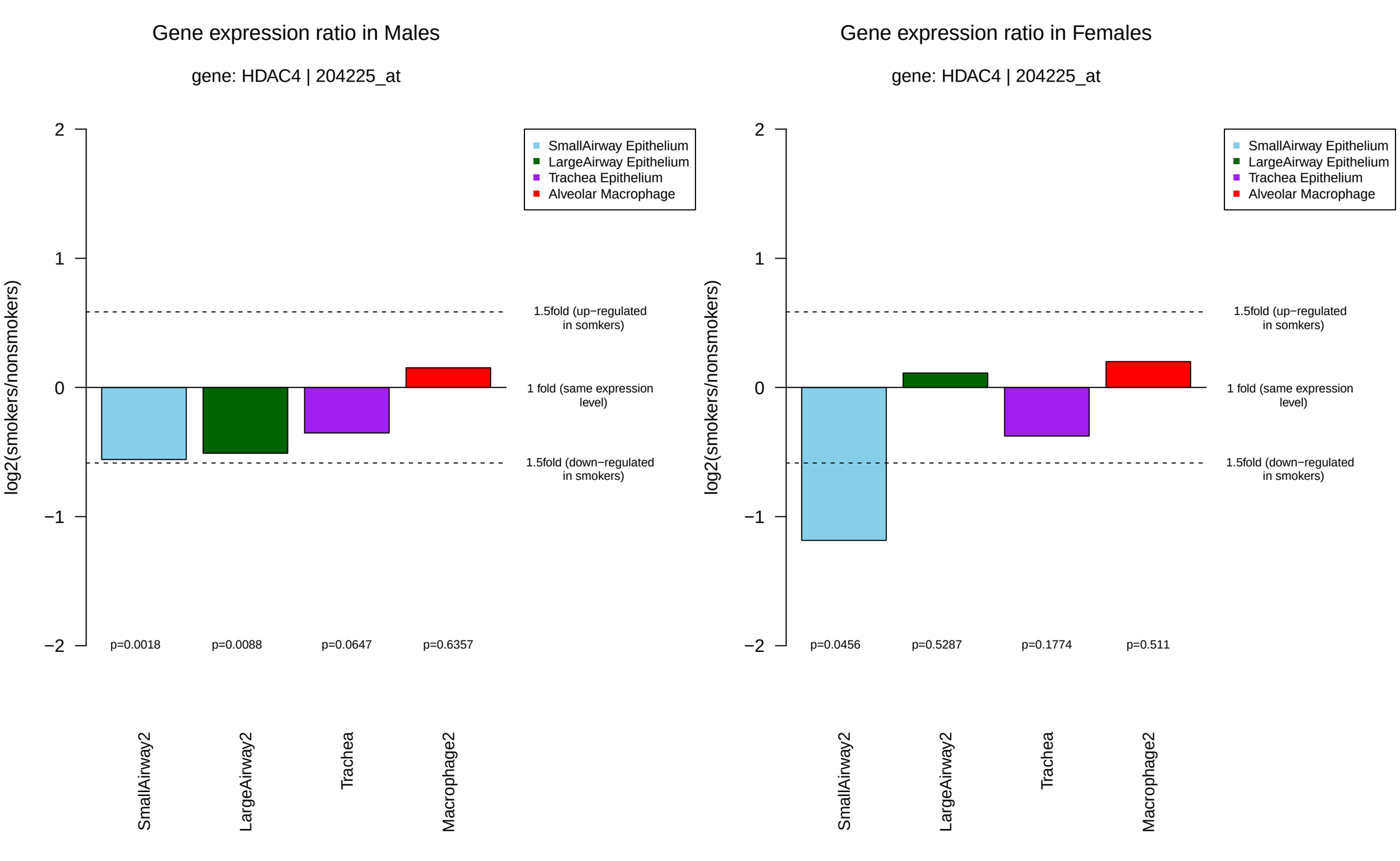
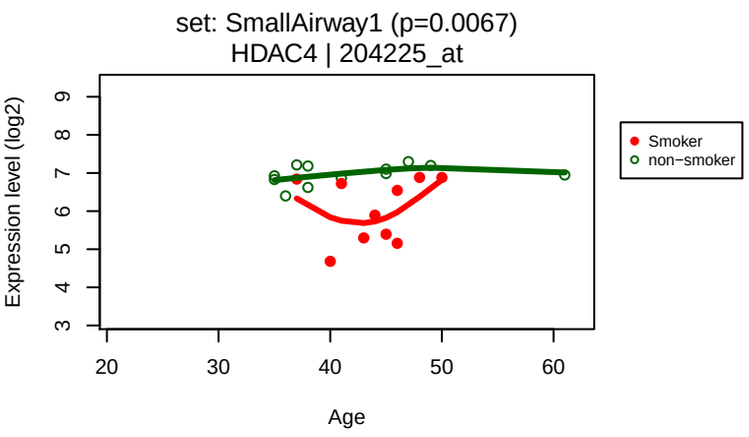
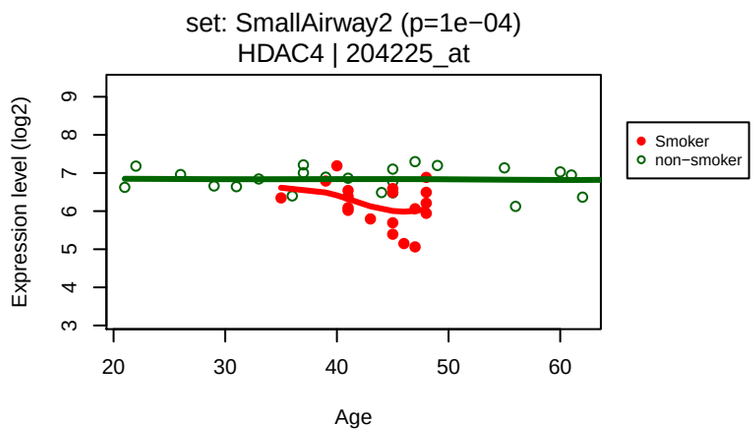
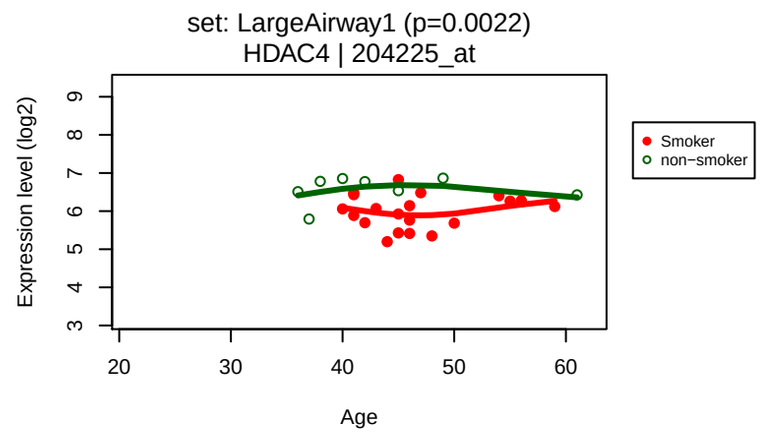

Gene expression of smokers and nonsmokers
gene: HDAC4 | 204225_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0067
p=1e−04
p=0
p=0.0022
p=0.0358
p=0.0329
p=0.8257
p=0.4477
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

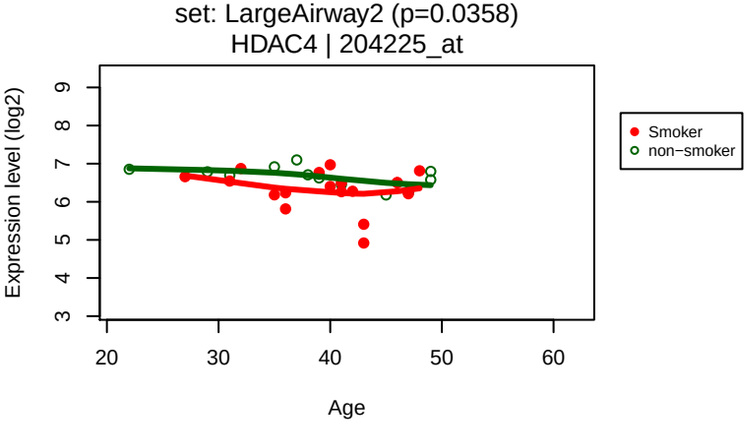

Gene expression ratio (smokers/nonsmokers)
gene: HDAC4 | 204225_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0067
p=1e−04
p=0
p=0.0022
p=0.0358
p=0.0329
p=0.8257
p=0.4477
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

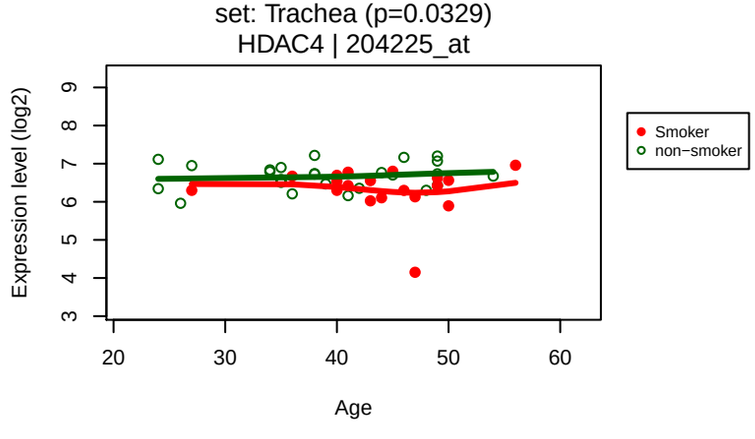

Gene expression ratio in Males
gene: HDAC4 | 204225_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0018
p=0.0088
p=0.0647
p=0.6357
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HDAC4 | 204225_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0456
p=0.5287
p=0.1774
p=0.511
SmallAirway2
LargeAirway2
Trachea
Macrophage2

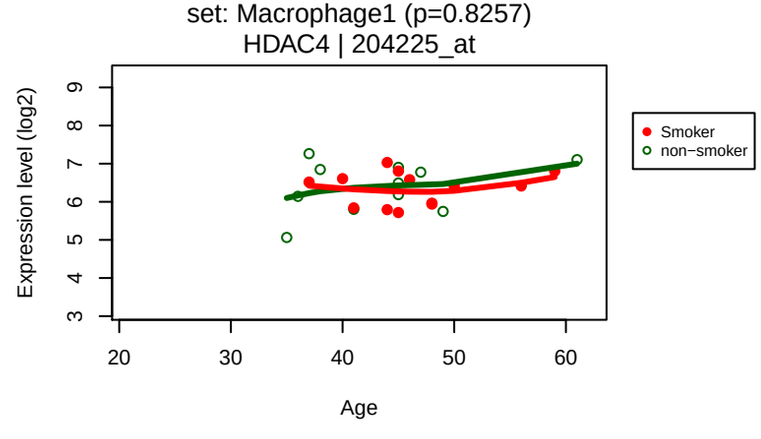

set: SmallAirway1 (p=0.0067)
HDAC4 | 204225_at
Expression level (log2)
Age
Smoker
non−smoker

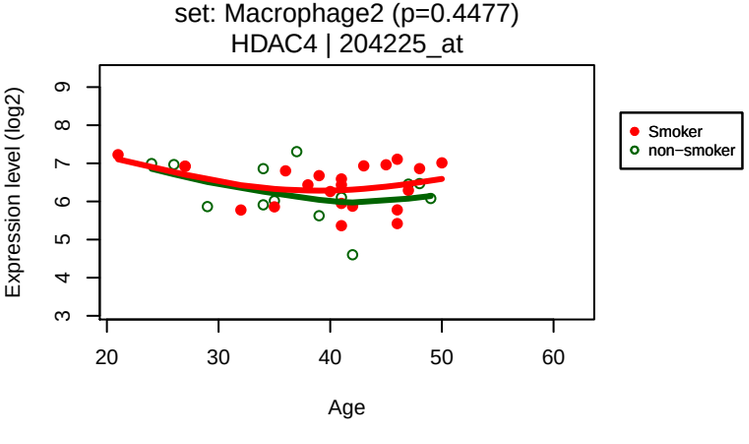

set: SmallAirway2 (p=1e−04)
HDAC4 | 204225_at
Expression level (log2)
Age
Smoker
non−smoker

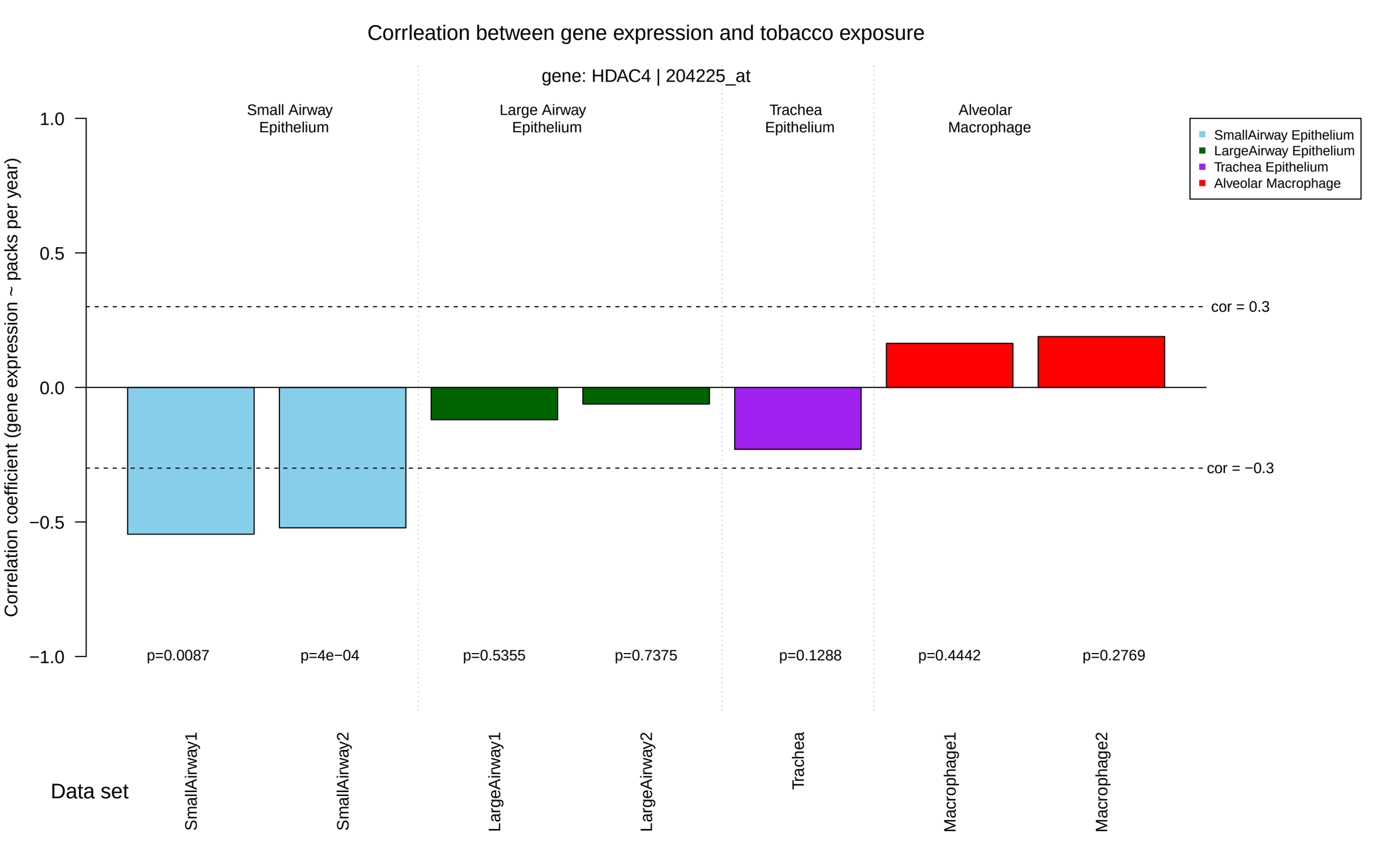

set: LargeAirway1 (p=0.0022)
HDAC4 | 204225_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0358)
HDAC4 | 204225_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0329)
HDAC4 | 204225_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8257)
HDAC4 | 204225_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4477)
HDAC4 | 204225_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HDAC4 | 204225_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0087
p=4e−04
p=0.5355
p=0.7375
p=0.1288
p=0.4442
p=0.2769
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2